

SCH3U

Test : Unit 4

Solutions & Solubility

Name: _____

Instructions:

- 1) You will have the entire period to complete the test (75 minutes)
- 2) The test is made up of 4 sections:
 - Section A – Multiple Choice / True and False
 - Section B – Short Answer
 - Section C – ESSAY QUESTIONS
 - Section D – PROBLEM BASED QUESTIONS
- 3) Please follow all the instructions
- 4) Please read all questions carefully
- 5) A mark breakdown of the test can be found after these instructions
- 6) Good Luck!

Mark Summary

Section	Knowledge /Understanding	Thinking	Application	Communication	TOTAL
A	10	/	/	/	10
B	/	2	4	4	10
C	/	4	/	4	8
D	11	14	/	/	25
Sub-Totals	/ 21	/ 20	/ 4	/ 8	/ 53

Comments:

Part A MULTIPLE CHOICE [K/U] (10 marks)

Circle the most suitable answer for the following questions (each question – 1 mark):

1. What is a solute?

- a) solution that contain water
- b) a substance that is dissolved in a solvent
- c) greatest amount in a solution

d) a homogeneous mixture

2. Anti freeze liquids are mixed with water in car radiators to form solutions with low freezing points, what type of solution is this?

- a) solid dissolved in liquid
- b) liquid dissolved in gas
- c) liquid dissolved in liquid
- d) liquid dissolved in solid
- e) a& d

3. To make up a solution with a specific volume, chemist use a piece of special glassware called _____?

- a) a beaker
- b) a wash bottle
- c) a volumetric flask
- d) a container

4. What rule should be followed when diluting concentrated acids

- a) is to add water to acid
- b) is to add acid to water
- c) a & b

5. What is the symbol of molar concentration?

- a) M
- b) C
- c) m
- d) c
- e) V

6. What is Solubility?

- a) property of solution
- b) the concentration of solution at given temperature
- c) property of solute
- d) the concentration of saturated solution at given temperature
- e) **a and b**
- f) **b and c**
- g) **a and d**

7. What factors affect solubility?

- a) size of crystals
- b) pressure
- c) shaking
- d) temperature
- e) **a and b**
- f) **b and d**
- g) **c and d**

8. How solubility of gasses depends on temperature?

- a) increase with temperature
- b) decrease with temperature
- c) doesn't depend on temperature
- d) prediction is not reliable

9. How solubility of solids depends on temperature?

- a) increase with temperature
- b) decrease with temperature
- c) doesn't depend on temperature
- d) prediction is not reliable

10. How solubility of liquids depends on temperature?

- a) increase with temperature
- b) decrease with temperature
- c) doesn't depend on temperature
- d) prediction is not reliable

Part B SHORT ANSWERS: (10 marks)

Give a short answer in 2-3 sentences to the following situational questions. (2 marks for each question).

1. Give three examples of aqueous solutions: one with a solid solute, one with a liquid solute, and one with a gas solute? [T/I:2]

2. Name two pieces of glassware that are used by chemists to prepare a solution with a known concentration? [C:2]

3. When you open a can of pop, which is more likely to fizz and spray: a can at room temperature or a cold can from refrigerator? Explain why using your understanding of gas solubility. [App:2]

4. Some Industries, particularly electric power generating stations, get rid of waste water by releasing it into a nearby lake or river. Use your knowledge of solubility and temperature to describe why this thermal pollution is detrimental to most fish. [App:2]

5. How does dilution affect the pH of (a) acidic solutions? (b) basic solutions? [C:2]

Part C ESSAY QUESTIONS: (8marks)

1. Describe the technique used by the chemist and non-scientist use for grinding solids and how the factor like “surface area” effects the rate of dissolving? [T/I:4 C:4]

Part D PROBLEM BASED QUESTIONS: (25 marks)

1. How would you prepare 400 mL of a 3.5 % (m/v) solution of sodium acetate? [K/U:4]
2. Gasoline sold in Ontario must contain at least 5.0% v/v ethanol. How much ethanol is a driver likely to get when buying 30 L of gasoline? [K/U:5]
3. Water is added to 100mL of 0.15 mol/L sodium nitrate NaNO_3 , to make 700 mL of diluted solution. Calculate the molar concentration of the diluted solution. [T/I:6]
4. The task is to obtain silver from 0.015 M solution of AgNO_3 . Given is 0.4 g of metal Cu. Would all silver in 1000 ml of 0.015 M solution of AgNO_3 precipitate with given amount of Cu. If not, how much silver will remains in solution and how many gram of Cu should we add to obtain all silver from 1L of given solution? [T/I:8]

5. Calculate amount (in g) of precipitate formed in reaction of 100 ml 0.1 M solution of $\text{Ca}(\text{NO}_3)_2$ salt with 3.0 g Na_2SO_4 . [K/U:2]

